

Monocular SLAM: validation against ground truth, then the pool data

Gregory German, September 2026. Full report, code and every number: github.com/gregorygerman777-hub/sonar_sim, [SLAM/validation/REPORT.md](#)

Validation. A keyframe based monocular SLAM (initialisation, map, triangulation, local and global bundle adjustment) was built and checked on data with ground truth. On a rendered scene resembling the pool with exact ground truth it recovers a 10 m trajectory to 0.6 cm (ORB) and 0.2 cm (SIFT), and its 3D map lies within a median 0.9 cm / 0.5 cm of the true surface. On real underwater video (AQUALOC) it is under 1 cm while tracking but loses track, and on the TUM benchmark it is 2 to 4 times less accurate than COLMAP (no loop closure). Ground truth exposed two scale bugs, both fixed with regression tests.

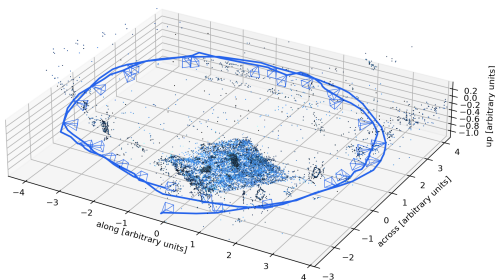
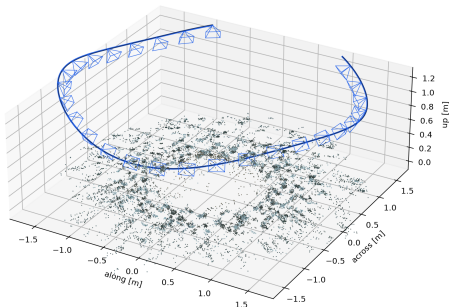
Pool data. The sequential SLAM holds only about 10 consecutive frames. COLMAP registers all 117 frames into one model (0.94 px reprojection error): the camera circles the rock target about twice at constant height, and frames 106 to 117, taken 9.6 minutes later, continue on the same circle. Where both methods pose the same frames, their relative rotations agree to within 1 degree. There is no ground truth, so this is plausible and internally consistent, not verified. The scale is unknown (monocular).

Likely cause of the pool tracking failures. Stills about 2 s apart with moving sunlight caustics. In the synthetic scene, adding moving caustics makes tracking fail at 5 degrees between frames, where the static scene works at 15 degrees.

Dataset	Method	Frames posed	ATE RMSE	ATE / path	Map error (median)
Synthetic pool (exact GT)	COLMAP (seq.)	240/240	0.003 m	0.03 %	0.008 m
Synthetic pool (exact GT)	ours (ORB)	240/240	0.006 m	0.06 %	0.009 m
Synthetic pool (exact GT)	ours (SIFT)	240/240	0.002 m	0.02 %	0.005 m
Synthetic pool + caustics	COLMAP (seq.)	240/240	0.005 m	0.05 %	0.013 m
Synthetic pool + caustics	ours (ORB)	240/240	0.019 m	0.19 %	0.024 m
Synthetic pool + caustics	ours (SIFT)	240/240	0.007 m	0.07 %	0.012 m
TUM fr1/xyz	COLMAP (seq.)	798/798	0.009 m	0.11 %	
TUM fr1/xyz	ours (ORB)	798/798	0.039 m	0.49 %	
TUM fr1/xyz	ours (SIFT)	798/798	0.022 m	0.27 %	
TUM fr3/long office	COLMAP (seq.), 1 in 3	862/862	0.021 m	0.09 %	
TUM fr3/long office	ours (ORB)	2585/2585	0.058 m	0.26 %	
TUM fr3/long office	ours (SIFT)	2585/2585	0.073 m	0.33 %	
AQUALOC harbor 07	COLMAP (seq.), 1 in 2	482/1131	0.006 m	0.06 %	
AQUALOC harbor 07	ours (ORB)	947/2261	0.008 m	0.10 %	
AQUALOC harbor 07	ours (SIFT)	502/2261	0.005 m	0.14 %	
Pool (117 stills)	COLMAP (exh.)	117/117	no ground truth		
Pool (117 stills)	ours (ORB)	10/117	no ground truth		
Pool (117 stills)	ours (SIFT)	11/117	no ground truth		

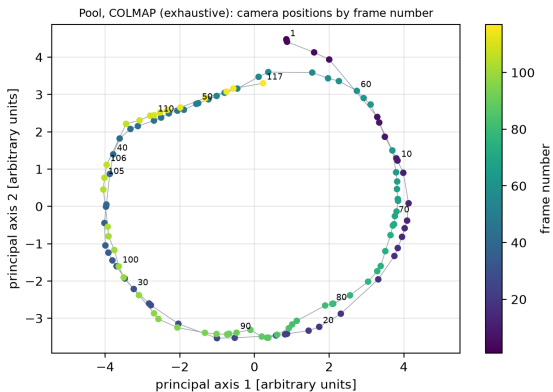
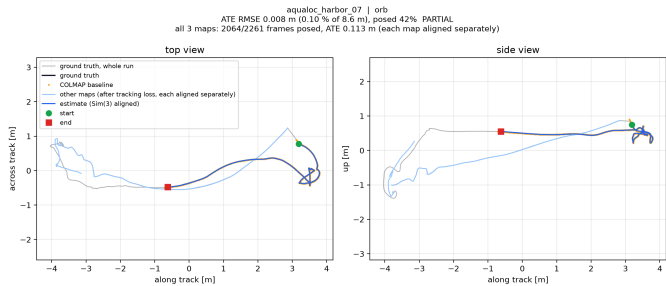
synthetic_pool | orb
trajectory over the 3D map: 15948 points shown (16479 in map), ATE RMSE 0.006 m (0.06 % of 10.0 m), posed 100%
— estimated trajectory
— ground truth

pool_width_scaled | colmap_exhaustive
trajectory over the 3D map: 12040 points shown (12117 in map), no ground truth; posed 117/117 frames
— estimated trajectory



Synthetic pool (exact ground truth): estimated trajectory and camera frustums over the reconstructed 3D map; the ground truth trajectory is drawn but indistinguishable.

Pool data: COLMAP trajectory (all 117 frames) and camera frustums over the 3D model; the dense patch is the rock and pebble target. Arbitrary units.



AQUALOC harbor 07 (real underwater): main map 0.8 cm ATE over 8.6 m; light blue: Pool data: camera positions coloured by frame number. Frames 1 to 50 and 50 to 105 each make one lap; 106 to 117 (after a 9.6 min gap) continue on the same circle.

ATE: RMSE after Sim(3) alignment, on the frames each method posed (read it with the posed fraction). COLMAP got the same undistorted images and fixed intrinsics.